Summary

- To apply methodologies in taught week to SoI

Elements and Methodologies

Elements

- Requirements #reqt
- Subsystems #SS
- Design Problems
  - Design Objectives #DO
    - Targets/Constraints
    - *Informed by #reqt*
  - Design Variables #DV
    - Initial Ranges
    - *Informed by #physics limits*
    - *Individual #SS authority*
  - Design Enablers
    - Physical Models
    - *Mapping #DV to #DO*

Methodologies

- Requirement Ranking
  - By Dependencies
    - *act A* `<<precedes>>` *act B*
    - *=> #reqt A > #reqt B*
    - $100
    - **Matrix Method**
- Design Space Visualisation
  - Sensitivity Analysis
  - Parallel Plots
  - Response Surfaces/Heatmapping
- Axiomatic Design
  - #DV and #DO Coupling
  - Decoupling
  - Order of priorities for design space exploration
- QFD
  - Recuesive HOQ methodology
  - Map #reqt $\rightarrow$ #DO $\rightarrow$ #DV $\rightarrow$ Derived #DV
- Objective Feasibility
  - Single-Objective Feasibility
  - Multi-Objective Feasibility
    - Multi-Objective, Multi-Variate Orthotopic Design Space
- Designing for Optimisation/Robustness
  - Heuristic Search

    * Point based solutions
    * Pareto Front Optimisation
    * Push to constraints
  - Constraint-Driven Design
    * Start with constraint
    * Push initial orthotope to constraint
    * Manipulate size and shape for design parameter weighting

Outputs

- SysML model
  - #reqt set
    * Derived #reqt
  - Contraint Bocks
  - Parametric Diagram
  - Recursed parametrics
- Python Scripts
  - Implemented Design Enablers
  - Design Sapce Visualisations
  - Implementation of rstool